Drawing Straight Line Graphs

Exam code: 0607

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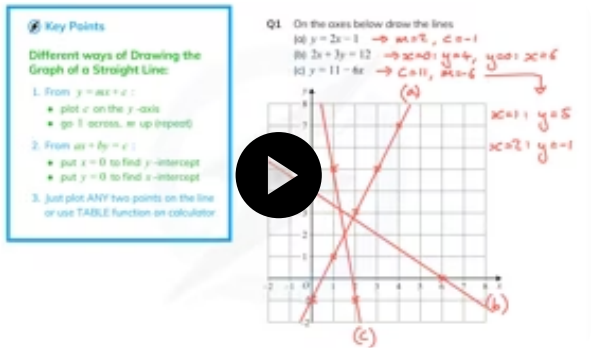

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Drawing linear graphs

How do I draw a straight line from a table of values?

- You may be given a **table of values** with **no** equation
- Use the *x* and *y* values to form a point with **coordinates** (*x*, *y*)
  - Then **plot** these points
  - Use a ruler to draw a **straight line** through them
    - All points should lie on the line
- For example
  - The points below are (-3, 0), (-2, 2), ... etc

x	-3	-2	-1	0	1	2	3
y	0	2	4	6	8	10	12

How do I draw a straight line using y = mx + c?

- Use the **equation** to create your own **table of values**
  - Choose points that are **spread out** across the axes given
- For example, plot $y = 2x + 1$ on axes from $x = 0$ to $x = 10$

Help me understand
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

x	0	5	10
y	1	11	21

How do I draw a straight line without using a table of values?

- Assuming the equation is in the form ***y = mx + c***
- **Start** at the ***y*-intercept**, *c*
- Then, for every **1 unit** to the **right**, go up ***m* units**
  - *m* is the gradient
  - If *m* is **negative**, go **down**
  - If *m* is a **fraction**, remember that gradient is change in *y* divided by change in *x*
    - A gradient of $\frac{a}{b}$ would be a units up for every b units right
- This creates a sequence of **points** which you can then join up
  - Be **careful** of counting squares if axes have **different scales**
    - 1 unit might not be 1 square

What if the equation is not in the form y = mx + c?

- Equations will not always be presented in the form ***y = mx + c***
- **Rearranging** to *y* = *mx* + c will make **plotting** these graphs **easier**
- Consider the equation $3x + 5y = 30$
  - Subtract 3*x* from both sides
    - $5y = -3x + 30$
  - Divide both sides by 5
    - $y = -\frac{3}{5}x + 6$
  - It can now be seen that the gradient is $-\frac{3}{5}$ and the *y*-intercept is 6
- Make sure you only have 1 *y* on one side, rather than say, 5*y*

How can I plot equations in the form ax + by = c?

- Instead of rearranging, equations in the form ***ax + by = c***, like the example above, can also be plotted by **considering the *x* and *y* intercepts** instead
  - Substitute in ***x* = 0** to find the *y*-intercept
  - Substitute in ***y* = 0** to find the *x*-intercept
  - E.g. for $3x + 5y = 30$
    - When $x = 0$, $5y = 30$, so $y = 6$
    - When $y = 0$, $3x = 30$, so $x = 10$

Examiner Tips and Tricks

Always plot at least 3 points (just in case one of your end points is wrong!)

Worked Example

On the same set of axes, draw the graphs of $\frac{y+1}{3} = x$ and $y = -\frac{3}{5}x + 3$.

Answer:

Rearrange $\frac{y+1}{3} = x$ into the form $y = mx + c$ to make it easier to plot

$$\frac{y+1}{3} = x$$
$$y + 1 = 3x$$
$$y = 3x - 1$$

For $y = 3x - 1$, create a table of values

x	0	1	2
y	-1	2	5

Plot the points (0, -1), (1, 2) and (2, 5)
Connect with a straight line

Alternatively, start at the y-intercept (0, -1) and mark the next points 3 units up for every 1 unit to the right

For $y = -\frac{3}{5}x + 3$, create a table of values
Because of the fraction, $x = 5$ is a good point to include

x	0	3	5
y	3	1.2	0

Plot the points (0, 3), (3, 1.2) and (5, 0)
Connect with a straight line

Alternatively, start at the y-intercept (0, 3) and mark the next points 3 units down for every 5 units to the right

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